NMA League Tables

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Introduction

The league table is the standard summary display for a network meta-analysis (NMA). It packs every pairwise treatment contrast — direct, indirect, and mixed — into a single triangular or square grid, making the relative ranking of all interventions visible at a glance. Each diagonal cell labels a treatment; each off-diagonal cell reports the pooled contrast between row and column with its 95 % confidence (or credible) interval. League tables are now a standard reporting requirement in the PRISMA-NMA extension because they communicate the entire network's information in a way that no single forest plot can.

Prerequisites

A working understanding of network meta-analysis (transitivity, consistency), how mixed treatment-comparison estimates combine direct and indirect evidence, and the conventions of pairwise effect sizes (OR, RR, SMD).

Theory

For T treatments the league table is a $T \times T$ matrix. The convention used by `netmeta` and most software places the random- or common-effect mixed estimates in the upper triangle and direct-only pooled estimates in the lower triangle, with the same treatment ordering on both axes. Each off-diagonal cell reports the pooled contrast (row-vs-column or column-vs-row, with the convention noted in the caption) and its uncertainty.

Ordering treatments by their rank (e.g., SUCRA, P-score, or median ranking from a Bayesian NMA) groups the best-performing treatments in one corner, making the table substantially easier to interpret than alphabetical ordering.

Assumptions

The underlying NMA is valid: transitivity holds across studies that compare different treatment subsets, consistency holds between direct and indirect evidence, and all pairwise contrasts are estimable from the connected network. Reporting league tables for a disconnected network is meaningless.

R Implementation

```r
library(netmeta)

set.seed(2026)
net_df <- data.frame(
  study = c("S1", "S1", "S2", "S2", "S3", "S3", "S4", "S4",
            "S5", "S5", "S6", "S6"),
  treat = c("A", "B", "A", "C", "B", "C", "A", "D",
            "B", "D", "C", "D"),
  TE    = c(0, -0.3, 0, -0.5, 0.4, 0.2, 0, -0.2,
            0.1, 0.3, 0, 0.4),
  seTE  = rep(0.25, 12)
)
```

```r
p_df <- pairwise(treat = treat, TE = TE, seTE = seTE,
                 studlab = study, data = net_df)
net  <- netmeta(p_df, common = FALSE, reference = "A")

league <- netleague(net, bracket = "(",
                    separator = " to ", digits = 2)
league$random
```

Output & Results

`netleague()` returns a structured matrix of pairwise contrast estimates with their confidence intervals, separately for the common-effect and random-effects models. The diagonal carries treatment labels; off-diagonal cells contain the pooled contrast and its CI in a formatted string. Exporting the table to LaTeX or HTML for publication uses standard formatting tools.

Interpretation

A reporting sentence: "The league table summarises all six pairwise contrasts in the network of four treatments. Treatment C was consistently the most effective: C vs A had a pooled effect of −0.55 (95 % CI −0.88 to −0.22) and C vs B was −0.34 (95 % CI −0.67 to −0.01); both contrasts excluded the null. The lower-triangle direct-only estimates were qualitatively concordant, suggesting consistency of direct and indirect evidence." Always pair league-table reading with a node-splitting consistency check.

Practical Tips

- Order treatments by their NMA-derived ranking (e.g., SUCRA from highest to lowest); the best-vs-worst contrast then sits in one corner and visual scanning becomes intuitive.
- Report both the point estimate and the CI; a significant contrast is one whose CI excludes the null on the analysis scale, but readers benefit from seeing the magnitude as well.
- For binary outcomes, exponentiate to OR or RR for clinical interpretation; on the multiplicative scale, a CI excluding 1 is the significance criterion.
- Complement the league table with a forest plot of contrasts versus a clinically meaningful reference treatment; the two views together communicate more than either alone.
- For networks with more than ten treatments, a full league table becomes unwieldy; consider a heatmap visualisation or restrict the table to clinically relevant pairs.
- Always cite the consistency assumption in the caption and report a node-splitting or design-by-treatment interaction check elsewhere in the manuscript.

R Packages Used

`netmeta::netleague()` for the canonical league-table output and `netmeta()` for the underlying frequentist NMA; `gemtc` and `multinma` for Bayesian NMA league-table equivalents using rank tables; `pcnetmeta` for component network meta-analysis with extended league-table summaries; `dmetar` for combined NMA workflow and reporting helpers.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Systematic reviews and PRISMA
- Meta-analysis basics
- Network meta-analysis